

ARTIFICIAL INTELLIGENCE × QUANTUM SCIENCE

Quantum Computing with Artificial Intelligence

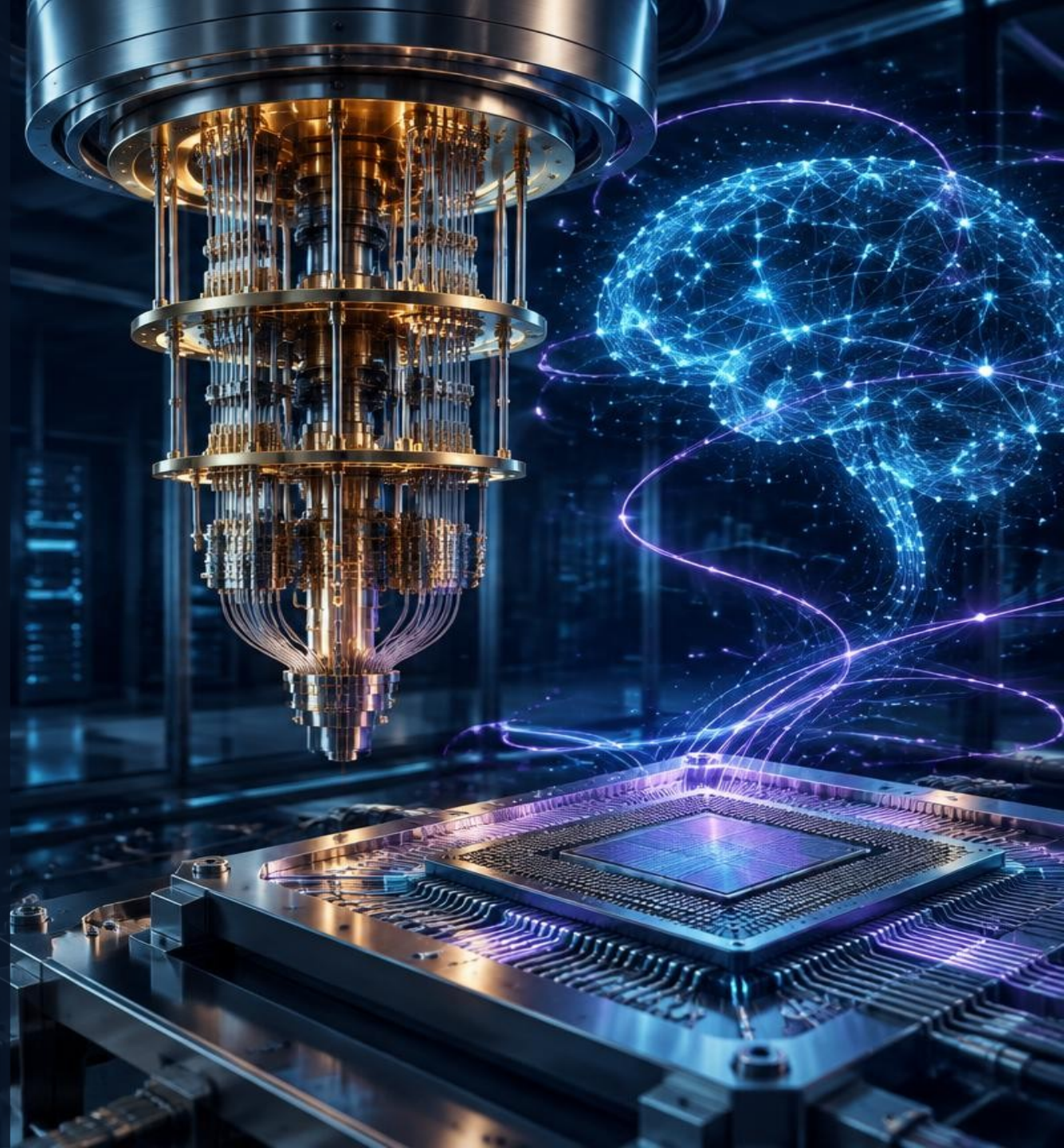
Exploring the future of intelligent technology

[Your Name]

Artificial Intelligence and Data Science

[Your College Name]

A COLLEGE PRESENTATION



Two powerful technologies

One learns from data. The other explores a different way to compute.

ARTIFICIAL INTELLIGENCE

Learns from examples

Finds patterns in information and uses them to make a prediction or response.

Like a helpful assistant that improves by studying examples.

+

QUANTUM COMPUTING

Uses quantum physics

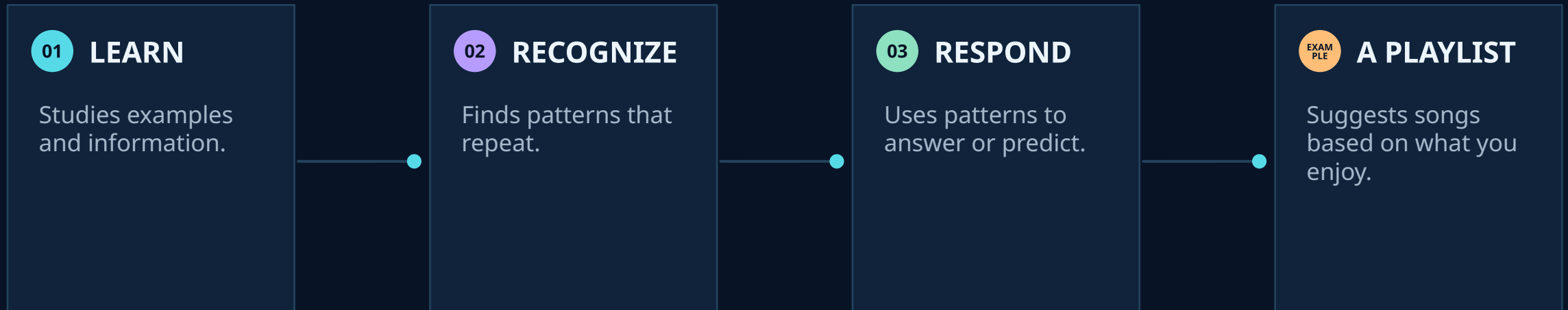
Processes information using quantum bits, designed for certain specialized problems.

Like a new kind of lab instrument—not a faster tool for everything.

Researchers are exploring how they could work together.

How does AI work?

A simple loop: examples help a system spot patterns and respond.



Everyday examples Voice assistants · photo recognition · movie recommendations · chatbots

What makes quantum computers different?

The key idea: quantum bits follow rules that ordinary bits do not.

TODAY'S COMPUTERS

bit

0 —  1

A bit is 0 or 1.

Like a switch set to OFF or ON.

QUANTUM COMPUTERS

qubit



Superposition

A combination of possible states

Entanglement

A special link between qubits

Quantum computers can be useful for some problems—not every task.

Two different ways to compute

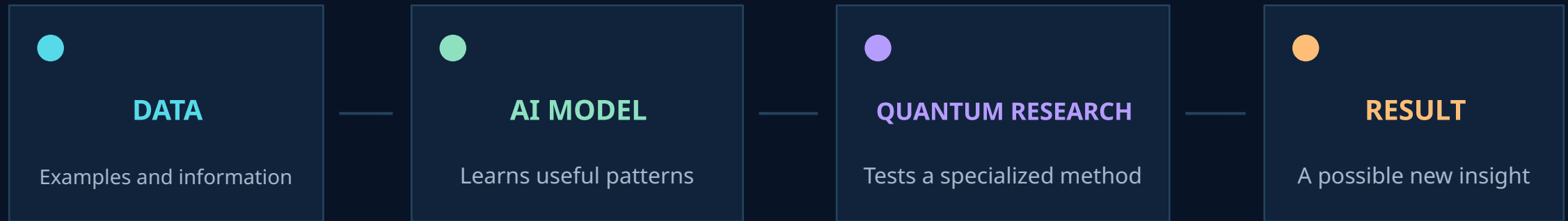
Quantum computers complement classical machines; they do not replace them.

CLASSICAL COMPUTING		QUANTUM COMPUTING	
Building block	Bits: 0s and 1s	Qubits: quantum states	
Best fit	Most everyday tasks	Some specialized problems	
Examples	Laptops, phones, web	Research and experiments	
Trade-off	Mature and practical	Sensitive, error-prone, developing	

Keep classical computers for daily work; use quantum machines where they may offer an advantage.

How could quantum computing help AI?

Researchers are testing whether quantum methods could assist selected steps.



CURRENT AI	in use today	QUANTUM-AI	active research; benefits not guaranteed
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Think of quantum computing as a specialized tool under investigation—not a magic speed boost.

Where could quantum AI be useful?

These are promising research directions—not established benefits everywhere.

01

HEALTHCARE

Explore molecules that matter for new medicines.

02

CLIMATE & ENERGY

Study complex systems and energy materials.

03

FINANCE

Investigate portfolio choices and risk.

04

TRANSPORT

Study difficult routing and scheduling.

05

MATERIALS & SCIENCE

Model matter to explore new materials.

Potential applications will need evidence, useful hardware, and fair comparisons.

Can it help medical research?

A possible future use: studying molecules to support drug-discovery research.

01

Research team asks a question

02

AI looks for patterns in data

03

Quantum simulation may test selected molecules

04

Scientists check results in the lab



Research support—not diagnosis, a guaranteed cure, or a replacement for clinicians.

What could we gain?

The opportunity is new problem-solving tools, if experiments prove they work.

Scientific insight

Ask new questions about complex systems.

New materials

Investigate molecules and matter.

Optimization

Explore selected hard planning problems.

New research paths

Create opportunities across science and technology.



QUANTUM
+ AI

Potential gains depend on the problem, hardware and method.

What are the challenges?

The technology is exciting, but it is still difficult to build and use well.



COST

Specialized machines and facilities are expensive.



STABILITY

Qubits are easily disturbed by their surroundings.



ERRORS

Noise can disrupt a computation; error management is hard.



SCALE & ACCESS

Large, reliable machines remain a technical challenge.



UNCERTAIN ADVANTAGE

Quantum methods do not automatically improve AI.

For most everyday AI today, conventional computers remain the practical choice.

What could the future look like?

Research is progressing; students can build foundations one step at a time.

- 1 AI fundamentals
- 2 Basic quantum concepts
- 3 Quantum programming
- 4 **Research projects**

POSSIBLE DIRECTIONS

AI engineering · computational science · quantum research

Specialized roles may call for deeper maths, physics and computer science.



THE TAKEAWAY

Building tomorrow with quantum computing and AI

- AI learns from data and makes predictions.
- Quantum computers use quantum physics.
- Together, they may help with selected problems.
- Research will show where they are truly useful.

"AI helps us learn from data. Quantum computing may open new ways to explore the problems of tomorrow."

THANK YOU

Questions & discussion

